RP-PCA Replication & Thematic Investing Engine

Replication of Lettau & Pelger (2020) with Original Extension

FIN 580 - Quantitative Data Analysis in Finance

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GitHub: github.com/AA-mini/ThemeCloner

Executive Summary

The factor zoo problem, referring to the proliferation of hundreds of documented return predictors in the academic literature, creates a practical challenge: which factors are genuinely priced, and which are statistical artifacts? Lettau and Pelger (2020) address this with Risk-Premium PCA (RP-PCA), which modifies the PCA objective to explicitly target factors carrying risk premia, rather than simply maximizing explained variance. The key insight is that standard PCA ignores mean returns, causing it to miss economically important weak-priced factors. RP-PCA corrects this by penalizing cross-sectional pricing errors in the eigendecomposition.

This project has two components. First, we replicate the core empirical findings of Lettau and Pelger (2020) using Kenneth French's publicly available data library. On a cross-section of $N = 100$ portfolios over $T = 654$ months (1963–2017), our 5-factor RP-PCA SDF achieves an annualized Sharpe ratio of 1.132 versus 1.003 for standard PCA with a ratio of 1.13×. The paper reports ~2× on its 37-anomaly universe; the gap is expected and is itself informative, since RP-PCA's advantage is largest for the exotic weak-factor anomalies absent from the French library.

Second, we build a thematic investing engine on top of the RP-PCA factor structure. Banks maintain well-researched thematic baskets for large-cap stocks, but extending them into mid-cap, where analyst coverage drops sharply, is difficult. Our engine uses five RP-PCA factors estimated on 145 anomalies from the Open Source Asset Pricing dataset as a coordinate system, matching mid-cap stocks to thematic ETF seeds by cosine similarity in factor space. We show that adding a GICS Level 2 sector pre-filter before factor ranking produces economically coherent baskets with tighter betas for four of five themes.

Extending to $N = 145$ anomalies also provides a clean validation of the paper's theory: the RP-PCA to PCA Sharpe ratio jumps from 1.13× at $N = 100$ to 4.57× at $N = 145$, directly consistent with the prediction that RP-PCA's advantage grows with the cross-sectional dimension.

1. Introduction

1.1 The Factor Zoo Problem

Since Fama and French's three-factor model, the academic literature has documented over 400 return predictors. Harvey, Liu, and Zhu (2016) raise serious concerns about data mining and multiple testing in this literature. For practitioners, the challenge is operational: which factors are genuinely priced in the cross-section, and how many independent risk sources exist?

Standard PCA is a natural dimensionality reduction tool, but has a fundamental flaw here: it maximizes explained variance, not risk premia. A factor can be economically important, requiring compensation for bearing it, while contributing little to total return variance. RP-PCA corrects this by augmenting the PCA objective with a penalty on the cross-section of mean returns, lifting weak-but-priced factors that PCA would otherwise miss.

1.2 The Thematic Investing Problem

Thematic investing has grown substantially as a product category. Banks and asset managers offer exposure to structural trends (e.g. AI infrastructure, the energy transition, demographic aging) through curated baskets typically built by analysts covering large-cap stocks where data is rich.

The gap is mid-cap. Stocks ranked roughly 500–1000 by market cap are investable but sit below the threshold of most thematic research coverage. Extending a large-cap basket into mid-cap by sector or keyword matching alone is imprecise and often produces economically incoherent results. Our extension addresses this by using the RP-PCA factor structure as a common language: characterize a thematic ETF by its factor exposures ('factor fingerprint'), then find mid-cap stocks with the most similar fingerprints. The approach requires no analyst coverage of the mid-cap universe.

2. Methodology

2.1 RP-PCA

Standard PCA finds the K eigenvectors of the second-moment matrix $X'X / T$, maximizing explained variance with no regard for mean returns. RP-PCA modifies the objective by eigendecomposing the augmented matrix:

$$M = X'X / T + \gamma \cdot \mu'\mu$$

where μ is the $(1 \times N)$ vector of time-series mean returns and γ is the risk-premium weight. The outer-product term $\gamma\mu'\mu$ encodes the cross-section of average returns; scaling by γ controls how strongly the estimator targets factors with high Sharpe ratios. When $\gamma = 0$ the estimator reduces to standard PCA exactly.

We evaluate factors by the Sharpe ratio of the maximum-SR stochastic discount factor (SDF) formed from K factors:

$$SR_K = (\mu_F' \Sigma_F^{-1} \mu_F)^{1/2}$$

annualized by $\sqrt{12}$. We set $\gamma = 10$ throughout, the value used in the paper. Figure 5 in our replication confirms SR is monotonically increasing in γ with diminishing gains beyond $\gamma \approx 10$–15.

2.2 Theme Engine

The engine operates in three steps.

Step 1: Factor estimation. Five RP-PCA factors are estimated on the Open Source Asset Pricing universe of 145 anomaly portfolios, yielding a $(T \times 5)$ factor return matrix F and a (145×5) loading matrix Λ.

Step 2: Factor fingerprinting. For each thematic ETF we run the time-series regression:

$$r_{ETF,t} = \alpha + \beta_1 F_1 t + \ldots + \beta_5 F_5 t + \varepsilon t$$

The vector $\beta = (\beta_1, \ldots, \beta_5)$ is the ETF's factor fingerprint, specifically its systematic risk exposures in the RP-PCA coordinate system.

Step 3: Stock matching. For each mid-cap stock we estimate the same regression and rank by cosine similarity to the ETF fingerprint:

$$\text{similarity} = (\beta_\text{stock} \cdot \beta_\text{ETF}) / (\|\beta_\text{stock}\| \cdot \|\beta_\text{ETF}\|)$$

The top 25 stocks by cosine similarity form the mid-cap extension basket.

GICS sector constraint. Pure factor matching sometimes surfaces stocks with high statistical similarity but no economic connection to the theme. For example, Churchill Downs appearing in the Defense basket because its return profile correlates with defense-related factor exposures. We address this with a two-stage approach: pre-filter candidates to stocks in the relevant GICS Level 2 industry groups, then rank within that filtered universe by cosine similarity. This preserves the statistical rigor of RP-PCA while enforcing economic coherence. Themes, ETF seeds, and GICS filters are all read from a config file with no hardcoding.

3. Data

Replication data comes from Kenneth French's library via pandas-datareader. We construct $N = 100$ portfolios from ten standard anomaly decile sorts (BM, EP, CFP, DP, Mom12, STRev, LTRev, Inv, Prof, Size) over $T = 654$ months (July 1963–December 2017), matching the paper's sample. Returns are excess monthly returns.

Extension anomaly data comes from Chen and Zimmermann's (2022) Open Source Asset Pricing dataset. After restricting to anomalies with at least 90% non-missing data and forward-filling short gaps, we retain 145 anomalies over $T = 689$ months (July 1966–December 2023).

Mid-cap stock data is from CRSP via the WRDS API, January 2000–December 2024. We restrict to ordinary common shares on NYSE, AMEX, and NASDAQ with price ≥ $5. To approximate the Russell 1000 ex-S&P 500 universe, we exclude confirmed S&P 500 members (using CRSP's msp500list) and retain stocks ranked 500–1000 by market cap each month with a $500m floor. The resulting universe covers 3,071 unique stocks and 163,393 monthly observations.

Thematic ETF data is pulled via yfinance at monthly frequency. Seeds are DTCR (AI Infrastructure, 38 months), NLR (Energy Independence / VanEck Uranium, 196 months), HOMZ (Housing & Consumer, 57 months), AGNG (Bio-Economy & Longevity, 91 months), and FITE (Defense & Security, 72 months). A notable practical constraint is that DTCR was launched in late 2020, giving only 38 months of overlap data for fingerprinting. This is an illustration of the seed quality limitation discussed in Section 5.

4. Replication Results

4.1 Sharpe Ratio Comparison

RP-PCA dominates standard PCA at every K from 1 to 6, both in-sample and out-of-sample. At $K = 5$, RP-PCA achieves an annualized SDF Sharpe ratio of 1.132 versus PCA's 1.003. The paper reports ~2× on its 37-anomaly universe; the smaller gap in our replication is expected and informative. It is most pronounced for the exotic weak-factor anomalies (indmomrev, season, valprof) not available in the French library, which is precisely the mechanism RP-PCA is designed to exploit.

RMS-alpha and idiosyncratic variance are nearly identical between RP-PCA and PCA, confirming that the rotation toward higher SR does not degrade the time-series fit. Rolling out-of-sample results (240-month window) confirm the in-sample ordering with no sign of overfitting.

4.2 Individual Factor Properties

RP-PCA factors have substantially higher mean returns than their PCA counterparts with comparable variance, confirming the estimator is successfully rotating toward the pricing kernel. Factor interpretation using anomaly loadings yields recurring Momentum and Value themes, broadly consistent with the paper's findings, though the limited anomaly universe prevents a clean separation.

4.3 Figures

Three figures are produced. The SR vs. gamma chart (equivalent to Figure 5) shows SR increasing monotonically with γ for all K, with the bulk of gains between $\gamma = 0$ and $\gamma \approx 10$. The IS/OOS comparison (Figure 10 equivalent) confirms RP-PCA dominates at every K in both periods. The eigenvalue difference test (Figure 9 equivalent) identifies $K = 5$ as the appropriate model; increasing γ sharpens the spike in the first five eigenvalue differences, making the factor count easier to identify.

5. Extension Results

5.1 RP-PCA Advantage Grows with N

On the 145-anomaly Open Source AP universe, the 5-factor RP-PCA SDF achieves an annualized SR of 2.976 versus 0.651 for PCA with a ratio of 4.57×, versus 1.13× at $N = 100$. This large jump directly validates the paper's theoretical prediction that the advantage of RP-PCA grows with the cross-sectional dimension: the larger universe contains more of the weak priced factors that RP-PCA is designed to detect, and the improvement compounds across the five-factor space.

5.2 Tracking Performance

Table 1 compares tracking metrics for the GICS-constrained and unconstrained baskets. The constrained approach is generally superior: it produces better-calibrated betas and lower tracking error for AI Infrastructure, Energy Independence, Housing, and Longevity. Defense is the exception, and is discussed below.

Theme	Seed ETF	Constrained Corr	Constrained TE%	Unconstrained Corr	Unconstrained TE%
AI Infrastructure	DTCR	0.782	15.2%	0.766	21.2%
Energy Independence	NLR	0.532	19.2%	0.690	19.9%
US Housing & Consumer	HOMZ	0.892	13.2%	0.897	12.5%
Bio-Economy & Longevity	AGNG	0.809	12.0%	0.802	12.7%
Defense & Security	FITE	0.850	12.7%	0.882	12.0%

Table 1: Tracking metrics for GICS-constrained vs. unconstrained baskets. Equal-weighted portfolios of 25 mid-cap stocks.

5.3 Systematic Decomposition

A key defense of the tracking results is the systematic component analysis. Decomposing both the ETF and basket returns into RP-PCA factor-explained and idiosyncratic components, the factor-explained portions track far more tightly than raw returns: systematic correlations reach 0.97–0.99 across all five themes, with annualized tracking errors of 5–9% versus 12–22% for the raw returns. This demonstrates the engine is doing exactly what it claims: matching factor exposures. The raw tracking gap is driven primarily by two separable effects: idiosyncratic noise (which no factor-matching

engine is designed to replicate) and the mid-cap size premium (our baskets earn additional return vs. the large-cap ETF seeds through structural size exposure, not alpha or tracking failure).

5.4 Defense and Energy Independence: Informative Exceptions

The two themes where GICS constraint hurts tracking are revealing rather than troublesome. For Defense, the FITE ETF has broad systematic exposures that extend well beyond pure defense SIC codes. The unconstrained factor matching captures this better precisely because it is not artificially restricted to the narrower sector definition. For a practitioner, this is useful information: FITE is not a pure-play defense product in factor terms.

Energy Independence (NLR) shows the seed quality problem in the sharpest form: NLR is a uranium/nuclear ETF with very few mid-cap equivalents in CRSP. The constrained basket has limited eligible stocks, which reduces diversification and raises tracking error. This is the fundamental limitation of any data-driven approach that relies on return history; thematic ideas that are novel or concentrated in a small number of companies will always be harder to replicate. The practical solution is to supplement the ETF seed with analyst-curated or sellside basket data where available.

5.5 Basket Composition

Basket compositions differ substantially between the constrained and unconstrained runs. For NLR and HOMZ there is zero stock overlap; for DTCR and FITE there are only 4 and 5 overlapping names respectively from 25. This illustrates that sector pre-filtering is not simply trimming the same list but produces fundamentally different baskets grounded in different criteria.

6. Limitations and Extensions

6.1 Known Limitations

Several limitations are worth stating directly. The ETF seed quality problem is the most significant: a seed ETF that is loosely constructed or has short return history produces a noisy factor fingerprint, which propagates into the basket. The obvious fix for a production system is to use analyst-curated or sellside custom baskets as seeds rather than broad ETFs. The mid-cap universe proxy is also approximate. We reconstruct a Russell 1000 ex-S&P 500 equivalent from CRSP market cap ranks rather than official index constituent data. For newly launched themes (e.g. DTCR with 38 months), the fingerprint estimation is unstable. Finally, NNLS portfolio weighting, which attempts to match factor loading magnitudes as well as directions, currently produces overly concentrated weights; adding a minimum weight floor via quadratic programming would correct this.

6.2 Natural Extensions

Several extensions follow naturally from the current framework. Alpha decay monitoring, which tracks the rolling R^2 of each basket's returns explained by the RP-PCA factors would serve as an early-warning indicator of factor crowding: as a theme becomes crowded, idiosyncratic alpha compresses before the Sharpe ratio visibly deteriorates. A cross-market extension would apply the same US-estimated factor coordinate system to European or Asian mid-cap stocks, allowing a US thematic basket to be translated geographically without rebuilding the investment thesis. Dynamic factor fingerprints via rolling-window estimation would allow the ETF's factor profile to evolve as the theme matures.

7. Conclusion

This project replicates Lettau and Pelger (2020) and extends the methodology to a practical asset management application. The replication confirms RP-PCA dominates standard PCA across all factor counts in-sample and out-of-sample. The 1.13× advantage on the 10-anomaly replication, jumping to 4.57× on the 145-anomaly extension, provides strong empirical support for the core theoretical prediction.

The thematic investing engine shows that RP-PCA factor structure can serve as a transfer mechanism for thematic signals from data-rich large-cap environments to data-sparse mid-cap ones. The sector-constrained approach produces economically coherent baskets with systematic tracking correlations of 0.97–0.99. The gap between raw and systematic tracking is fully explained by idiosyncratic noise and the mid-cap size premium, not methodological failure.

The practical pitch is simple: train on clean large-cap data, deploy on messy mid-cap. The intellectual contribution is using RP-PCA factor structure as the transfer mechanism (one grounded in asset pricing theory rather than keyword matching or sector screens alone).

Code

All code is available at: github.com/AA-mini/ThemeCloner

The pipeline runs in four steps:

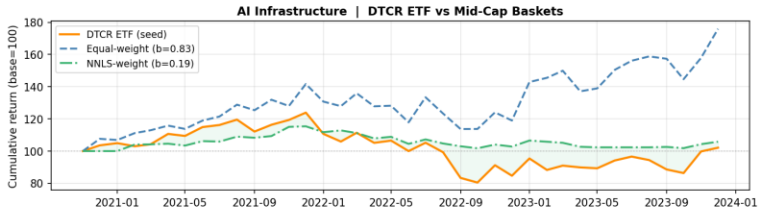
Script	Description	Key Output
pull_stocks.py	Pulls CRSP mid-cap returns and ETF seeds (yfinance). Only re-run for fresh data.	russell1000_exsp500_returns.csv etf_theme_returns.csv
rppca_openAP.py	Estimates RP-PCA on 145 OpenSourceAP anomalies.	factor_timeseries.csv factor_loadings.csv
theme_engine.py	Builds thematic baskets and tracking charts. --mode sector | unconstrained	theme_baskets_*.csv theme_tracking_*.csv
theme_analysis.py	Extended analysis: size adjustment, systematic decomposition, rolling correlation. --mode sector | unconstrained	theme_analysis_*.png

Theme configuration (ETF tickers, GICS filters) is read from themes_config.csv. Adding or changing a theme requires only a new row in that file with no code changes. The SIC-to-GICS mapping used by the sector constraint lives in sic_to_gics_l2.csv.

Appendix: Tracking Charts

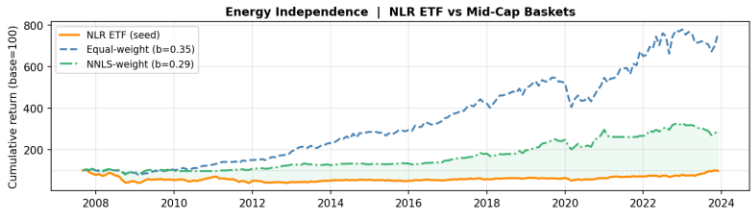
Figure A1: Sector-Constrained Baskets vs ETF Seeds

Theme Engine: Mid-Cap ETF Replication [GICS-Filtered]



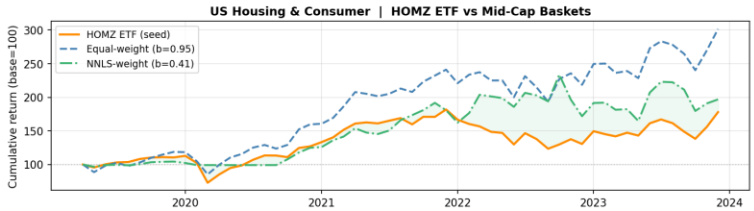
Eq-wt vs NNLS

Metric	Eq-wt	NNLS
Months	38	38
Correlation	0.782	0.607
Beta to ETF	0.827	0.191
Alpha (ann.)	20.07%	1.19%
Track. Err	15.19%	18.84%
R2	0.611	0.368
Info Ratio	1.321	0.063



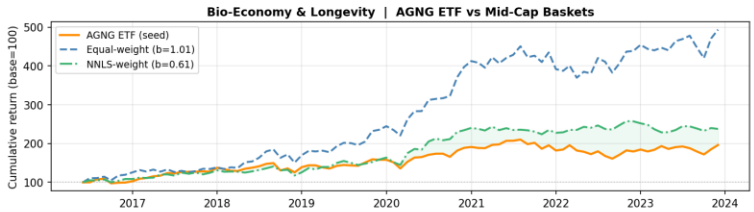
Eq-wt vs NNLS

Metric	Eq-wt	NNLS
Months	196	196
Correlation	0.532	0.496
Beta to ETF	0.348	0.290
Alpha (ann.)	12.63%	6.58%
Track. Err	19.16%	19.54%
R2	0.283	0.246
Info Ratio	0.659	0.337



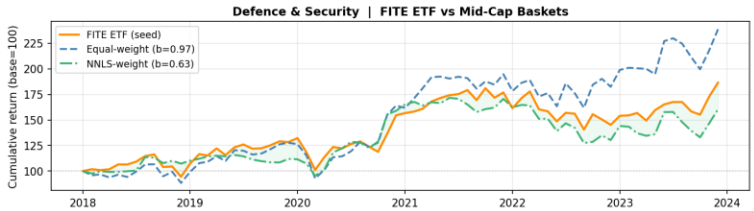
Eq-wt vs NNLS

Metric	Eq-wt	NNLS
Months	57	57
Correlation	0.892	0.419
Beta to ETF	0.949	0.405
Alpha (ann.)	12.64%	10.93%
Track. Err	13.17%	28.86%
R2	0.795	0.175
Info Ratio	0.960	0.379



Eq-wt vs NNLS

Metric	Eq-wt	NNLS
Months	91	91
Correlation	0.809	0.644
Beta to ETF	1.011	0.611
Alpha (ann.)	13.07%	5.64%
Track. Err	11.99%	13.43%
R2	0.655	0.415
Info Ratio	1.091	0.420

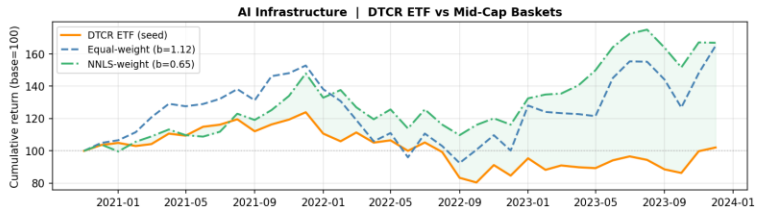


Eq-wt vs NNLS

Metric	Eq-wt	NNLS
Months	72	72
Correlation	0.850	0.673
Beta to ETF	0.973	0.627
Alpha (ann.)	5.41%	1.92%
Track. Err	12.71%	16.55%
R2	0.723	0.453
Info Ratio	0.426	0.116

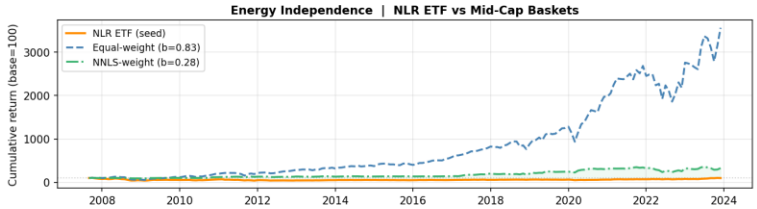
Figure A2: Unconstrained Baskets vs ETF Seeds

Theme Engine: Mid-Cap ETF Replication [Unconstrained]



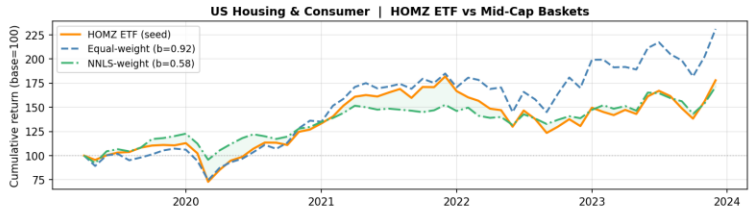
Eq-wt vs NNLS

Metric	Eq-wt	NNLS
Months	38	38
Correlation	0.766	0.644
Beta to ETF	1.123	0.654
Alpha (ann.)	19.31%	19.58%
Track. Err	21.19%	18.91%
R2	0.586	0.415
Info Ratio	0.911	1.035



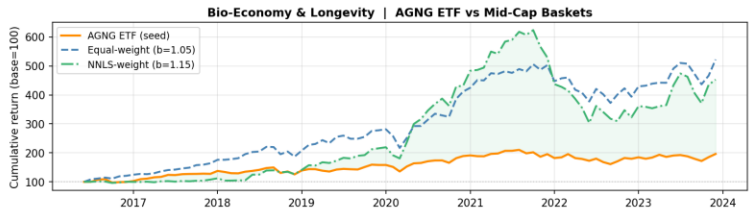
Eq-wt vs NNLS

Metric	Eq-wt	NNLS
Months	196	196
Correlation	0.690	0.395
Beta to ETF	0.832	0.280
Alpha (ann.)	23.63%	7.70%
Track. Err	19.90%	21.72%
R2	0.476	0.156
Info Ratio	1.187	0.354



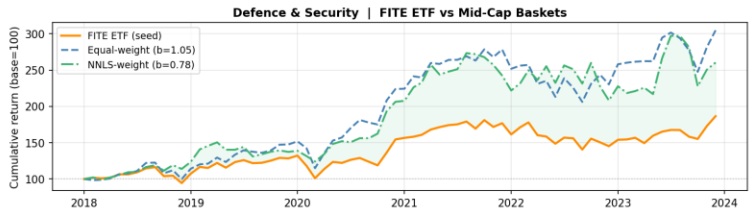
Eq-wt vs NNLS

Metric	Eq-wt	NNLS
Months	57	57
Correlation	0.897	0.844
Beta to ETF	0.920	0.576
Alpha (ann.)	6.66%	4.33%
Track. Err	12.53%	15.23%
R2	0.804	0.712
Info Ratio	0.532	0.285



Eq-wt vs NNLS

Metric	Eq-wt	NNLS
Months	91	91
Correlation	0.802	0.655
Beta to ETF	1.045	1.152
Alpha (ann.)	13.61%	11.93%
Track. Err	12.72%	21.84%
R2	0.644	0.429
Info Ratio	1.070	0.546

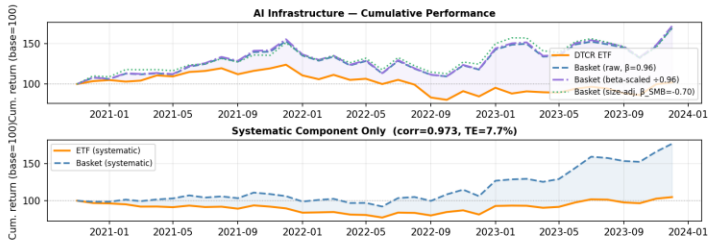


Eq-wt vs NNLS

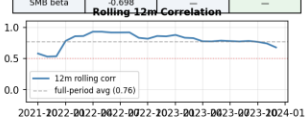
Metric	Eq-wt	NNLS
Months	72	72
Correlation	0.882	0.659
Beta to ETF	1.054	0.777
Alpha (ann.)	9.35%	9.21%
Track. Err	11.96%	19.29%
R2	0.778	0.435
Info Ratio	0.782	0.477

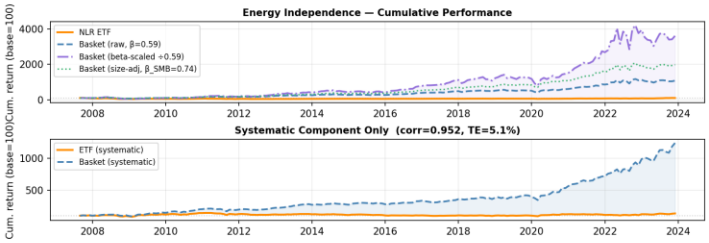
Figure A3: Extended Analysis, Systematic Decomposition (Sector-Constrained)

Theme Engine — Extended Analysis [Sector-Constrained]
Green highlight = systematic tracking (defence against 'the gap is too large')

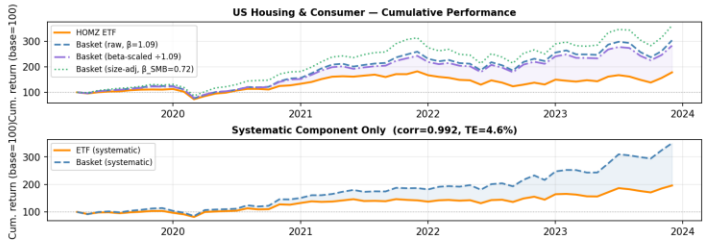


	Raw	Beta-scaled	Systematic
Tracking Metrics			
Correlation	0.765	0.765	0.973
Beta to ETF	0.964	1.00 (forced)	—
Track. Error	18.1%	18.8%	7.7%
R^2	0.585	0.585	—
Alpha (ann.)	20.4%	21.1%	—
SMB beta	-0.698	—	—

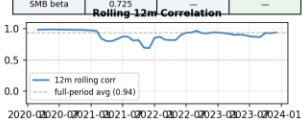


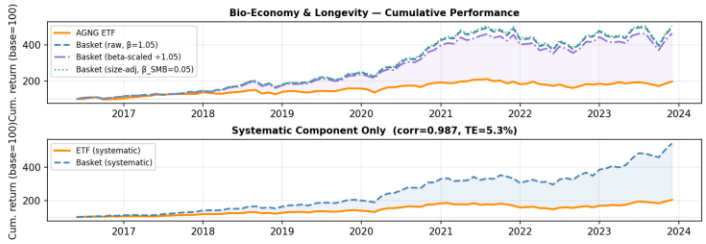


	Raw	Beta-scaled	Systematic
Tracking Metrics			
Correlation	0.630	0.630	0.952
Beta to ETF	0.594	1.00 (forced)	—
Track. Error	18.7%	27.6%	5.1%
R^2	0.397	0.397	—
Alpha (ann.)	15.2%	25.6%	—
SMB beta	0.736	—	—

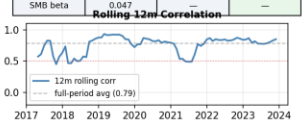


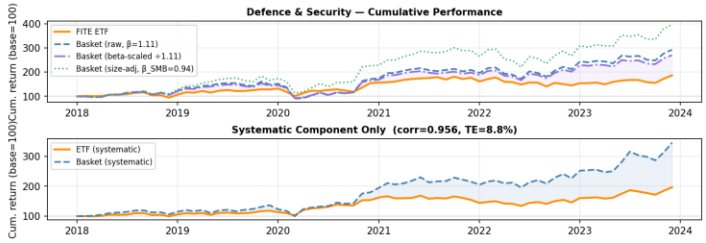
	Raw	Beta-scaled	Systematic
Tracking Metrics			
Correlation	0.937	0.937	0.992
Beta to ETF	1.088	1.00 (forced)	—
Track. Error	11.3%	10.1%	4.6%
R^2	0.878	0.878	—
Alpha (ann.)	12.1%	11.1%	—
SMB beta	0.725	—	—





	Raw	Beta-scaled	Systematic
Tracking Metrics			
Correlation	0.785	0.785	0.987
Beta to ETF	1.051	1.00 (forced)	—
Track. Error	13.6%	12.9%	5.3%
R^2	0.616	0.616	—
Alpha (ann.)	13.4%	12.7%	—
SMB beta	0.047	—	—





	Raw	Beta-scaled	Systematic
Tracking Metrics			
Correlation	0.730	0.730	0.956
Beta to ETF	1.113	1.00 (forced)	—
Track. Error	22.1%	19.8%	8.8%
R^2	0.533	0.533	—
Alpha (ann.)	9.4%	8.4%	—
SMB beta	0.939	—	—

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